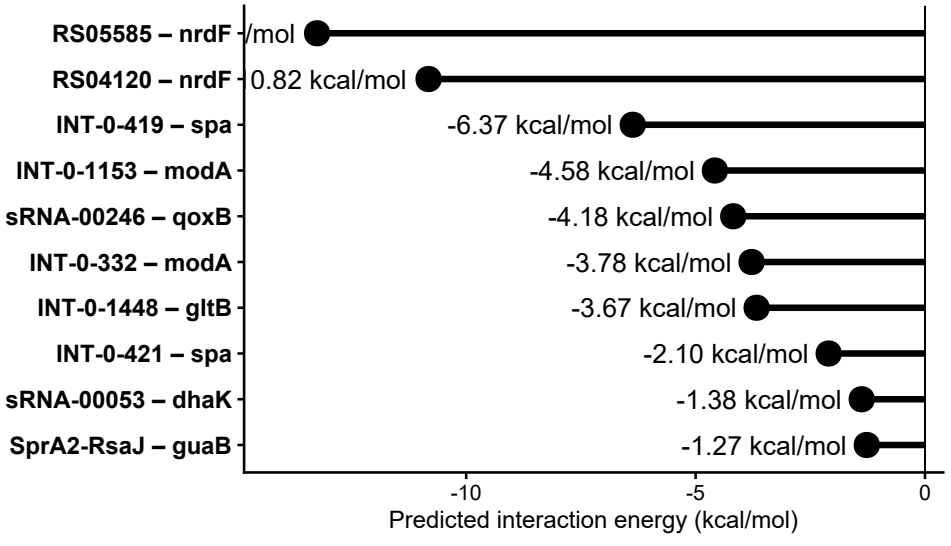
# IntaRNA-based structural characterization of CLASH-supported RNA–mRNA interactions

Predicted duplex energetics and binding-site localization are presented as structural descriptors rather than functional validation.

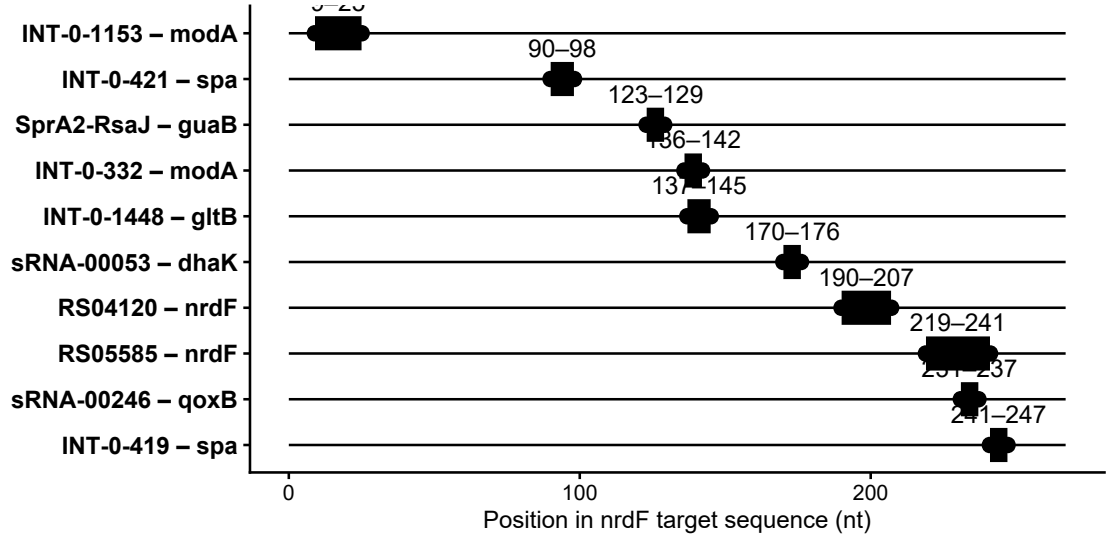
## A Predicted interaction energy

IntaRNA-predicted duplex energetics; more negative values indicate a more favorable predicted interaction



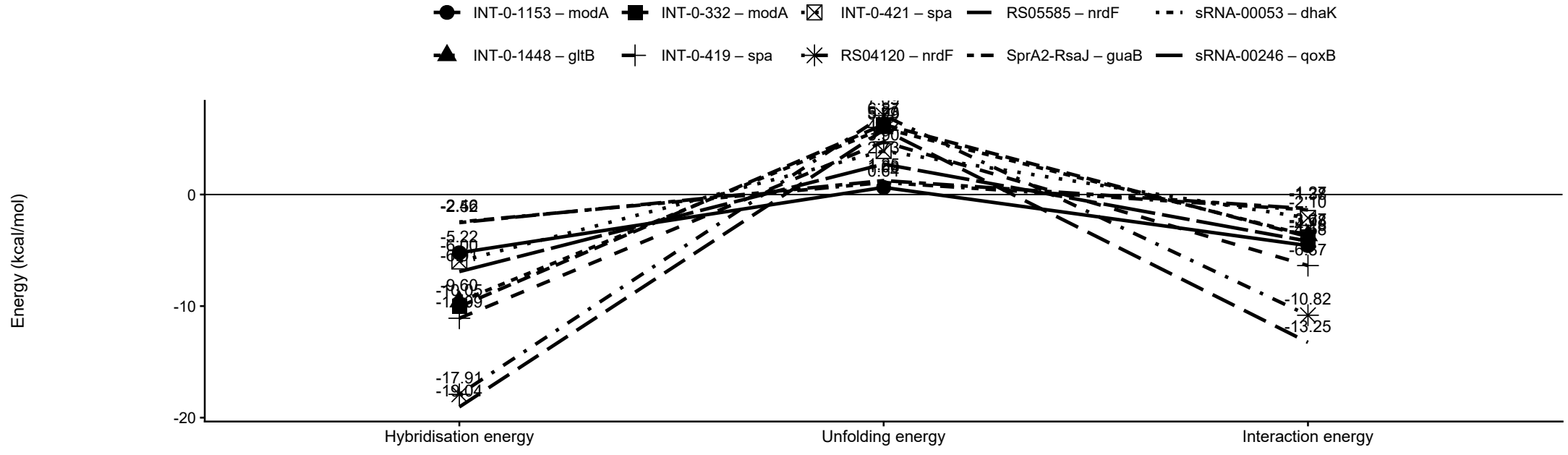

## B Predicted nrdF target-site localization

IntaRNA-predicted binding intervals on the target sequence; positions are reported from the IntaRNA i




## C Predicted energetic components

Hybridisation and unfolding contributions are shown separately from the resulting interaction energy




## D Structural prediction summary

Predicted properties are descriptive and are not treated as independent functional validation

| | RS05585 – nrdF | RS04120 – nrdF | INT-0-419 – spa | INT-0-1153 – modA | sRNA-00246 – qoxB | INT-0-332 – modA | INT-0-1448 – gltB | INT-0-421 – spa | sRNA-00053 – dhaK | SprA2-RsaJ – guaB |
|---|---|---|---|---|---|---|---|---|---|---|
| Interaction energy | -13.25 kcal/mol | -10.82 kcal/mol | -6.37 kcal/mol | -4.58 kcal/mol | -4.18 kcal/mol | -3.78 kcal/mol | -3.67 kcal/mol | -2.10 kcal/mol | -1.38 kcal/mol | -1.27 kcal/mol |
| Hybridisation energy | -19.04 kcal/mol | -17.91 kcal/mol | -11.09 kcal/mol | -5.22 kcal/mol | -6.91 kcal/mol | -10.05 kcal/mol | -9.60 kcal/mol | -6.00 kcal/mol | -2.46 kcal/mol | -2.52 kcal/mol |
| Unfolding energy | 5.79 kcal/mol | 7.09 kcal/mol | 4.72 kcal/mol | 0.64 kcal/mol | 2.73 kcal/mol | 6.27 kcal/mol | 5.93 kcal/mol | 3.90 kcal/mol | 1.08 kcal/mol | 1.25 kcal/mol |
| RNA-binding region | 7–27 | 2–17 | 76–82 | 1–17 | 17–23 | 26–32 | 6–14 | 40–48 | 1–7 | 7–13 |
| nrdF-binding region | 219–241 | 190–207 | 241–247 | 9–25 | 231–237 | 136–142 | 137–145 | 90–98 | 170–176 | 123–129 |
| Seed length | 7 nt | 7 nt | 7 nt | 7 nt | 7 nt | 7 nt | 7 nt | 7 nt | 7 nt | 7 nt |